

Response to JPE Replication Report (JPE-Jin-20210685-R1)

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Manuscript: *Buying Data from Consumers: The Impact of Monitoring in U.S. Auto Insurance*

Report date: April 23, 2026

Response date: May 19, 2026

We thank the Data Editor and the reproducibility team for the careful and constructive report. Below we address each item raised.

A. Glitches in execution (Report §8, §9.1)

A.1 `max_row` runtime error in CTF equilibrium grid

Resolved. `max_row` is now defined inline in `ctf_equi_k_save_grid.R` before first use. The pipeline runs to completion without intervention.

A.2 `doParallel` not declared as a dependency

Resolved. `doParallel` (and `foreach`, `data.table`, `tictoc`, `MASS`, `fixest`) are now listed in the README’s Section II package list and in the `install.packages()` block immediately below it.

B. Manuscript and README updates (Report §2 item 3, §5, §5.1, §6)

B.3 Data-provider confidentiality footnote

Added to both the manuscript title footnote (acknowledgments section) and the README’s Data Availability Statement: “The empirical analysis uses proprietary data from a major U.S. auto insurer provided under a confidential data use agreement. The identity of the data provider cannot be disclosed per the terms of the agreement; the *JPE* Data Editor has verified the existence of the data provider and the data use agreement under a separate non-disclosure agreement.”

The signed NDA was sent to the Data Editor, to be followed by the verification package.

Our DUA specified that we were not permitted to disclose state names, so throughout the paper, we now refer to Illinois, which constitutes our estimation sample, as the focal state.

B.4 Author affiliations in README

Added. University of Toronto and Stanford University now appear at the top of `README.md`, matching the manuscript title page.

B.5 Data citation (Report §5.1)

Added to both the manuscript reference list and the README. The proprietary panel and the Quadrant Information Services competitor price quotes each have formal `@misc` entries in `paper/literature.bib` and are cited at first mention in Section 2.

B.6 Codebook (Report §5)

Addressed. The proprietary data is not shipped, so a record-level codebook does not apply. The two distributional-profile JSON files in `data/simulated/` document every variable name, type, dimension, and distributional target used to generate the synthetic data. The README’s “Simulation profiles” subsection points researchers to these files for variable-level verification.

B.7 Data-preservation statement (Report §5)

Addressed. The Data Availability Statement now states explicitly that the proprietary panel will be preserved for at least five years after publication under the data use agreement, and that the authors will provide reasonable assistance to replication requests.

B.8 Computational requirements (Report §6.1)

Addressed. The README’s Computational Requirements section now includes per-task hardware guidance, a runtime/storage matrix for both cached and uncached Section II runs, and tested hardware specs.

C. Figure B.3 (Report §2 item 2; §9)

Resolved. We discovered a small bug that overwrote Figure B.3 during the robustness counterfactual runs; the current `paper.pdf` version is the correct one. This is the only exhibit change to the real-data (non-simulated) pipeline.

D. Missing appendix exhibits (Report §9.3)

Resolved. The original submission’s `run_simulated.R` produced 7 of 17 model-fit and CTF appendix exhibits. The simulated pipeline now generates all 47 exhibits with zero skips.

To speed up the Section II (simulated) verification run, we have shipped cached estimation and counterfactual outputs in `output/simulated/cached/*.tar.gz`. The cache is restored automatically when the simulated pipeline is invoked with its defaults:

```
Rscript code/run_simulated.R # uses cache (~15 minutes)
```

To run the simulated pipeline from cold (no cache restored, re-estimating every model and re-solving every counterfactual):

```
Rscript -e 'USE_CACHE <- FALSE; source("code/run_simulated.R")'
```

E. Items respectfully dismissed

E.1 PII flags (Report §7.1)

All flagged tokens are substrings of unrelated variable names (`lat` → `latent`, `son` → `Poisson`, `loc` → `block`, `zip` → an anonymized 3-digit prefix used in regression dummies). No individual-level records appear in any shipped file.

E.2 Windows / mixed paths (Report §7.1.2)

All “Windows path” and “mixed path” hits are LaTeX backslash commands inside R string literals (“`\\begin{tabular}`”, “`\\midrule`”, etc.). The single “drive letter” hit is “`config.R:\n`” inside a `cat()` call. No absolute paths, `setwd()` calls, or platform-specific references exist in the codebase.

E.3 macOS metadata / duplicates (Report §7.1.3)

The `__MACOSX/` directories and `.DS_Store` duplicates are Finder artifacts, not shipped files. They will be stripped from the archive prior to deposit on the Dataverse.

Sincerely, Yizhou Jin and Shoshana Vasserman